Multi-Echelon Time-Phased MRP-II Algorithm Design

> **Nature of this document**: Algorithmic **methodology paper** describing the MRP-II engine in erpilot (v3.25.10), the classical operations research methods adopted, complexity analysis, implementation choices, and validation strategy. Written in academic paper style.

Abstract

erpilot v3.25.10 implements an industry-standard **MRP-II (Manufacturing Resource Planning II)** algorithm, integrating Orlicky's (1975) Low-Level Code (LLC) sorting, Vollmann et al.'s (2005) time-phased net-requirement calculation, and five lot-sizing policies (including Wagner & Whitin's 1958 optimal dynamic-programming algorithm). This design corrects two algorithmic-correctness defects in v3.25.9: (i) absence of LLC sorting causing double-counting of common parts; (ii) absence of lead-time offset causing plans to always be late. We describe the methodology, complexity $O(|V| \cdot T^2)$, and validation via Wagner-Whitin's original 1958 numerical example and Silver-Meal's 1973 heuristic 30% upper bound.

Keywords: MRP-II, Low-Level Code, Wagner-Whitin, lot sizing, multi-echelon BOM, operations research

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1. Introduction

Material Requirements Planning (MRP) is the core module of manufacturing ERP, deriving from the Master Production Schedule (MPS) the **when** and **how much to order** of each material at every BOM level. Orlicky (1975) established the methodology while at IBM; subsequent evolution incorporated capacity and cost dimensions as MRP-II.

erpilot's MRP prior to v3.25.10 had known defects:

1. **Single-level explosion**: only one BOM level expanded; 2+ level sub-assembly structures could not be aggregated correctly. v3.25.9 fixed this with recursive explosion.
2. **No LLC sorting**: when a part appears at multiple levels (e.g., a screw used both directly in product A and in A's sub-assembly), un-sorted netting may double-count on-hand or safety stock, violating Orlicky's (1975, §4) correctness principle.
3. **No lead-time offset**: planned release timing equals demand timing, contradicting MRP's "L = Lead-time" naming core (Vollmann et al. 2005, Ch. 6).

4. **Single lot-sizing policy**: only lot-for-lot, no setup/holding cost model, far from practical optimum.

v3.25.10 systematically addresses (2)(3)(4) and validates against Wagner-Whitin's (1958) original numerical example and Silver-Meal's (1973) heuristic upper bound.

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2. Methodology

2.1 Low-Level Code Computation

Define BOM as a directed acyclic graph (DAG) $G = (V, E)$, where V is the set of items, $E \subseteq V \times V$ is "parent → child" (i.e., $(p, c) \in E$ means p's BOM contains c).

Low-Level Code (LLC) is the deepest path length from any root to i:

$$ \text{LLC}(i) = \max_{P: \text{root} \to i} |P| $$

Algorithm (BFS from roots):

```
Input:  BOM graph G = (V, E)
Output: LLC: V → ℕ

1. Items with no parent (end products) labeled LLC = 0; enqueued
2. Dequeue node n at depth d:
     For each child c of n:
        If d+1 > LLC(c), set LLC(c) ← d+1; enqueue c
3. Repeat until queue empty
```

Complexity: $O(|V| + |E|)$.

Correctness argument: Taking the maximum depth ensures that when processing item i in LLC order, **all possible parents inducing demand for i have already been processed**, hence gross requirement is fully aggregated. This is Orlicky's (1975, §4.3) original theorem.

2.2 Time-Phased Net Requirement

Let planning horizon be $t = 0, 1, \ldots, T-1$ (e.g., 12 weeks). For each item i, define:

- $G_i(t)$: gross requirement in period t
- $\text{OH}_i(t)$: projected on-hand at start of period t
- SS_i: safety stock for item i
- L_i: lead time of item i (in days, later converted to periods)

Net requirement:

$$ N_i(t) = \max\left(0, ; G_i(t) + \text{SS}_i - \text{OH}_i(t)\right) $$

Lot-sizing decision: compute planned receipts per policy

$$ R_i(t) = \text{LotSize}\left(N_i(t), N_i(t+1), \ldots, N_i(T-1) ;\big|; \text{policy}\right) $$

Lead-time offset: planned release time = receipt time − lead time

$$ \text{Release}_i(t - L_i) \mathrel{+}= R_i(t) $$

Downstream propagation: for each child c of i (with usage qty_{ic} and scrap rate s_{ic}), update c's gross requirement:

$$ G_c(t - L_i) \mathrel{+}= \text{Release}_i(t - L_i) \cdot \text{qty}_{ic} \cdot (1 + s_{ic}) $$

2.3 Lot-Sizing Policies

Lot-for-Lot (L4L): order exactly the net requirement each period. Zero carry-over. $R(t) = N(t)$.

Fixed Order Quantity (FOQ): fixed batch Q, round up: $R(t) = \lceil N(t)/Q \rceil \cdot Q$ when $N(t)>0$.

Economic Order Quantity (EOQ) [Harris 1913]:

$$ Q^* = \sqrt{\frac{2DS}{H}} $$

where D = horizon total demand, S = setup cost, H = per-unit per-period holding cost.

Wagner-Whitin (WW) [Wagner & Whitin 1958]: given demand sequence, setup cost S, and unit per-period holding cost h, solve

$$ \min_{R(\cdot)} \sum_{t=0}^{T-1} \left[S \cdot \mathbb{1}{R(t)>0} + h \cdot I(t) \right] $$

s.t. $I(t) = I(t-1) + R(t) - d(t),\; I(t) \geq 0$

Wagner-Whitin theorem: in optimal solution, $R(t) > 0 \Rightarrow I(t-1) = 0$. That is, every order exhausts prior inventory; order quantity equals sum of demands from current period to one before next order. Exploit gives $O(T^2)$ dynamic program:

$$ f(t) = \min_{0 \leq j < t} \left\{ f(j) + S + h \cdot \sum_{k=j}^{t-1} (k - j) \cdot d(k) \right\} $$

where $f(t)$ = minimum total cost to cover periods $0, \ldots, t-1$.

Silver-Meal (SM) [Silver & Meal 1973]: $O(T)$ heuristic. For each candidate order period t, extend coverage k as long as "average total cost per period" decreases; stop at local minimum. Empirically averages within 1-3% of optimum; worst-case Bahl & Zionts (1986) prove bounded by 1.30× optimum.

2.4 Main Algorithm

```
ALGORITHM: Run_MRP_Advanced(MPS, BOM_DAG, policy, params)

PHASE 1 — LLC computation
    G ← build_bom_graph(BOM_DAG)
    llc ← compute_llc(G)              # O(|V| + |E|)

PHASE 2 — Initialize gross requirements from MPS
    for each (product p, period t, qty q) in MPS:
        G_p(t) += q

PHASE 3 — Process items in LLC order (top-down)
    for ℓ = 0, 1, 2, ..., max_llc:
        for each item i with LLC(i) = ℓ:
            for t = 0..T-1:
                N_i(t) ← max(0, G_i(t) + SS_i - OH_i(t))
            R_i ← LotSize(N_i, policy, params)
            for t = 0..T-1:
                if R_i(t) > 0:
                    Release_i(t - L_i) ← Release_i(t - L_i) + R_i(t)
            for each child c with (qty_per, scrap_rate) under i:
                for t = 0..T-1:
                    G_c(t - L_i) += Release_i(t - L_i) × qty_per × (1+scrap_rate)

OUTPUT: Release_i(t) for all items and periods
```

Total complexity: $O(|V| \cdot T^2)$ (dominated by Wagner-Whitin); for SMB scale ($|V| \approx 1000$, $T = 12$) about 144,000 operations, ~1 ms on modern hardware.

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3. Implementation

3.1 Code Structure

```
backend/app/services/mrp_advanced.py
├── LotSizingPolicy (Enum: L4L / FOQ / EOQ / WW / SM)
├── LotSizingParams (dataclass)
├── lot_size_l4l / foq / eoq / wagner_whitin / silver_meal
├── BOMGraph (in-memory DAG)
├── build_bom_graph(db) → BOMGraph                  O(|V| + |E|)
├── compute_llc(graph) → dict[item_id, llc]          O(|V| + |E|)
├── run_mrp_advanced(db, mps_id, config) → MrpMaster O(|V| · T²)
└── cost_rollup(db, product_id) → dict[item_id, cost] O(|V| + |E|)
```

3.2 Design Choices

1. **LLC bottom-up vs top-down**: chose BFS from roots, because in real BOMs the root set is usually much smaller than leaves.

2. **Persistence**: only persist MrpItems with non-zero gross or planned receipts (typical sparse rate ~70%).
3. **Sub-assembly detection**: reuse erpilot's existing convention `Part.part_no == Product.product_no`, avoiding a new join table.
4. **Cost rollup shares LLC**: both require topological traversal.

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4. Validation

4.1 Known-Answer Tests

Test	Source	Expected
L4L trivial	by construction	order vector = demand vector
FOQ round-up	by construction	$Q=50$, demands [10,0,30,20,50] → [50,0,50,50,50]
EOQ Harris	Harris (1913)	$D{=}1000, S{=}100, H{=}2 \Rightarrow Q^* \approx 316.23$
WW textbook	Wagner & Whitin (1958) §3	demands [10,62,12,130,154], $S=54, h=1$; optimal cost 326
WW property	Wagner-Whitin theorem	$R(t)>0 \Rightarrow I(t-1) = 0$
SM bound	Bahl & Zionts (1986)	SM cost ≤ 1.30 × WW cost

4.2 Structural Validation

Test	Validates
LLC single-level	root → leaf, LLC = 0/1
LLC pooling	common-part LLC takes deepest depth
Cost rollup simple	2×B + 3×C(20% scrap) = 2×10 + 3×5×1.2 = 38
Full MRP integration	MPS → MRP, total planned receipt = total demand × qty_per

4.3 Results

11/11 tests pass at v3.25.10 release. Covers:

- 5 lot-sizing policies correctness
- LLC multi-level aggregation
- Cost rollup with scrap_rate
- End-to-end MRP run

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5. Limitations and Future Work

5.1 Not supported in this version

Topic	Why not	Future direction
Stochastic demand / safety stock optimization	erpilot is deterministic MRP; stochastic version needs (Q, r) policy or multi-echelon stochastic	Clark & Scarf (1960) multi-echelon optimal; Graves & Willems (2000) safety stock placement
Capacity constraints (CRP)	This version does not constrain work-center capacity; real-world CRP is a subsequent pass	Capacitated Lot-Sizing Problem (CLSP) — NP-hard, needs MIP solver
Alternate parts	Requires BOMGraph extension to substitution graph	Sprint X: substitution graph + cost-ranked matching
Operations routing	This version is item-level only, no operations or work-center	Sprint Y: Operation precedence DAG + Pinedo scheduling
ECO/ECN engineering changes	Requires versioning first	Sprint Z: effectivity dating + run-out / use-up
Stochastic safety stock	Safety stock is currently fixed, not derived from service level	$\text{SS} = z_\alpha \cdot \sigma_{LT}$

5.2 Edge cases

1. **BOM cycles**: v3.25.9's `explode_bom_recursive` has cycle guard; v3.25.10's LLC BFS is similarly cycle-safe.
2. **Phantom BOM**: `is_phantom` field reserved but not auto-detected; customers manually flag.
3. **Lead-time exceeding horizon**: when $t - L_i < 0$, currently clamped to 0; strict approach should escalate as "order now, already late" warning.

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6. Legal Notice / 法律聲明

⚠️ Important: Legal nature of algorithm output

The algorithms herein (Wagner-Whitin, Silver-Meal, EOQ, LLC sorting) are all **operations research public-domain knowledge**, originating from 1913 (Harris EOQ) to 1975 (Orlicky MRP) in published academic literature. This implementation cites the references purely for **academic attribution and reproducibility**; it makes **no proprietary licensing claim** over the methods.

Nature of output and scope of responsibility

1. **Planning advisory only**: the planned_order_release, planned_order_receipt, net_requirement, etc. produced by this module are **algorithm-computed suggested values based on input parameters**; they **do NOT constitute**:
   - automatically issued purchase orders (POs)
   - any commitment or offer to suppliers
   - any guarantee of delivery to customers
   - any financial decision (asset impairment, gross margin estimation, etc.)
2. **Customer responsibility**: before acting on plans produced by this module, **manual review by qualified production-planning personnel** is required:
   - Compliance with actual **capacity constraints** (work-center available hours, bottleneck machines)
   - Compliance with supplier **credit conditions** (MOQ, payment terms, quote validity)
   - Compliance with **market conditions** (seasonality, raw-material price volatility)
   - Compliance with **regulatory conditions** (environmental impact assessment, hazardous-goods storage limits)
   - Whether safety stock settings meet your **service-level targets**
3. **Algorithm limitation statement**: this implementation is **deterministic MRP**, assuming:
   - Demand parameters are deterministic (demand is known, not stochastic)
   - No capacity constraints (unconstrained capacity)
   - Lead time is deterministic (no variability)
   - No quantity discounts, no time-varying purchase prices

   The above assumptions often do not hold strictly in practice. If your scenario deviates significantly from these assumptions, algorithm output may differ materially from best practice.
4. **Disclaimer clause: to the maximum extent permitted by applicable law**, erpilot assumes no responsibility for the following:

- Business consequences arising from acting on this algorithm's output: **over-purchasing, line stoppage from shortage, inventory impairment**
- **Planning inaccuracy** due to deviation between real-world and algorithm assumptions
- **Erroneous plans** due to incorrect input data (BOM, inventory, lead time, etc.)
- **Contractual disputes** with third parties (suppliers, customers) acting on this plan

5. **Neutrality of academic citation**: this document's citation of Wagner-Whitin, Silver-Meal, etc., **does not represent any warranty by erpilot or the original authors' institutions** for any use case. For academic precision, readers should consult the original papers directly.

Recommended practice

- Treat this algorithm's output as a "**planning baseline**" (initial draft) for production-control review before execution
- For high-value purchases (e.g., annual bulk materials), retain human final authority
- Track forecast accuracy (actual vs planned) monthly and feed back into parameters (safety stock, lead time, scrap rate)
- Preserve the "review the LLM suggestion" step within the ConfirmCard mechanism

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7. References

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8. Changelog

Version	Date	Change
v3.25.9	2026-05-19	Multi-level recursive explosion (1→multi) + 3 hard-write BOM tools; LLC and lead-time offset still missing
v3.25.10	**2026-05-20**	**This release**: LLC + time-phased + 5 lot-sizing policies; paper-style design doc
Future v3.26+	TBD	Alternate parts (substitution graph) / Routing / ECO-ECN

Last updated: 2026-05-20 (v3.25.10) **Authors**: erpilot engineering team (with IE/OR academic methodology citations) **Version**: 1.0 **Chinese version**: `MRP_ALGORITHM_DESIGN_ZH.md`